

tagonist to escape the undesired outcome, and then adversarially establish the conditions that block the protagonist from that escape. **Detail Elaboration** §3.3 composes event sequences that manifest each of the planned actions and failures, then elaborates on the details of the events. Our method does not rely on any external story corpora. Once the problem is decomposed in this way, the LLM is capable of generating all intermediate information required for generating a full suspenseful story.

3.1 Background Setup

We first sample a genre (e.g., spy thriller) from a list of suspenseful story genres¹ obtained by prompting the LLM. Next, we ask the LLM for the name and occupation of the protagonist. Conditioned on that information, we prompt the LLM to design a goal for the protagonist to achieve. Based on the goal, we then use the LLM to generate a dire situation that the protagonist would be put in, if the protagonist couldn’t achieve the goal. After that, we feed all the information to the LLM and ask it to generate an introductory paragraph of a story that can cover the given information. The prompts we use in the background setup stage and an example of LLM’s corresponding outputs are presented in Table 1.

3.2 Outline Planning

Inspired by the psychological theory on suspense (Gerrig and Bernardo, 1994), we plan the story outline by iteratively creating story plots in which the protagonist takes actions trying to achieve the goal, followed by the failure of the actions. Specifically, we realize it by iteratively prompting the LLM to (step #1) generate a series of possible actions the protagonist could take to achieve the goal. For each action, we also prompt the LLM to (step #2) generate a potential reason that makes the action turn out to be ineffective in helping the protagonist achieve the goal—avoid the negative outcome—and get out of the dire situation. To maintain the story coherence, during prompting we make sure that each action is generated conditioned on the failure of all previous actions, and that the likelihood of the series of actions is decreasing—readers assume a rational protagonist will try the best options first. In this way, readers can start to feel suspense as they witness the gradual elimination of a series of feasible plans to avoid

Tell me about the name and the occupation of a protagonist of a spy thriller.

Max Sterling, a highly skilled and enigmatic operative for a clandestine government agency tasked with eliminating high-profile threats to global security.

(We denote the model output as \$name_occupation.)

Tell me about a very concrete goal the protagonist wants to achieve.

Max Sterling is determined to track down the elusive leader of an international terrorist organization known as “The Phoenix”, who is planning a devastating attack on a major world summit. Max’s goal is to gather enough evidence to dismantle the organization and bring their leader to justice before innocent lives are lost.

(We denote the model output as \$goal.)

Tell me about a dire situation that the protagonist would be put in, if the protagonist couldn’t achieve the goal.

If Max Sterling fails to achieve their goal, “The Phoenix” will successfully carry out their planned attack, resulting in the loss of countless lives and substantial damage to global security. With the organization’s leader remaining at large, they would continue orchestrating acts of terrorism, further destabilizing the world and posing an ongoing threat that would be difficult to contain. Max understands the urgency and stakes involved, knowing that failure could lead to catastrophic consequences.

(We denote the model output as \$dire_situation.)

Write an introductory paragraph of a story that covers all the above information.

In the shadows of a world teetering on the brink, an enigmatic operative named Max Sterling emerged from the depths of secrecy. Loyal to a clandestine government agency devoted to preserving global security, Max possessed an unmatched skillset and a steadfast determination. Their mission: to unearth and dismantle “The Phoenix”, an international terrorist organization poised to unleash chaos upon a major world summit. As the clock ticked ominously, Max embarked on a relentless pursuit, driven by the weight of countless innocent lives hanging in the balance. Failure would mean catastrophic consequences, with “The Phoenix” thriving and global security faltering. But Max, undeterred by the daunting odds, knew that the time for redemption had come with each heartbeat. The fate of nations rested upon their resilient shoulders, and the chase had only just begun.

Table 1: The prompts we use in subsection 3.1 Background Setup (highlighted in **bold**), and an example of LLM’s corresponding outputs (highlighted in *italic*). The genre of this story example is *spy thriller*.

the negative outcome while the negative outcome draws closer and time runs out.

The prompts we use in the outline planning stage and an example of LLM’s corresponding outputs are presented in Table 2. For the illustration purpose, in this example we only perform three iterations of prompting and get three pairs of actions and reasons; in the third iteration we only keep \$action3 and omit \$reason3 as we want the protagonist to succeed at the end of the story. In practice, there is no limitation on the number of

¹We provide the list in Appendix A.

prompting iterations, allowing our method to generate an unlimited length of stories with increased levels of storyline complexity.

3.3 Detail Elaboration

With the action-reason pairs obtained in subsection 3.2, we convert each pair into a short summary for a story chapter in which the protagonist tries to take the given action but that action does not work due to the given reason. Conditioned on the short summary, we prompt the LLM to generate a sequence of events that can serve as an outline for each story chapter. During prompting we need to make sure the protagonist is not aware that their plan will fail (readers assume the protagonist is rational and will not take actions that they believe are unlikely to be effective in achieving their goal). With a sequence of events, the full story is then generated by asking the LLM for an elaboration on the event details.

Aiming to further improve the level of suspense readers could possibly perceive, we incorporate two optional designs into our method for suspenseful story generation. (1) we ask the LLM to set up some small clues for the upcoming failure of the protagonist’s action so as to better arouse readers’ anticipation of the protagonist’s failure. (2) We draw inspiration from the narrative theory of *disparity of knowledge* (Branigan, 2013) and design two information revelation modes. In the first, we reveal the reason the protagonist’s actions will fail *beforehand*, setting up an anticipation of failure in the reader that the protagonist doesn’t know about and is thus unlikely to be able to avert. For example, if our protagonist is a spy, they might be unaware that the antagonists have set a trap. In the second, we reveal the information *after the fact*, explaining to the reader how the protagonist’s actions were doomed to fail from the beginning.

A complete run of our method starts with §3.1 Background Setup and may contain two or more prompting iterations. Each of the prompting iterations comprises §3.2 Outline Planning and §3.3 Detail Elaboration, which are repeated in our method to create stories in which the protagonist repeatedly fails to prevent the dire situation. In the final iteration, we do not generate the failure response (i.e., step #2 of subsection 3.2), and we allow the protagonist to succeed in preventing the dire situation and achieving the goal. The prompts we use in the detail elaboration stage and an example of LLM’s corresponding outputs are presented in Ap-

Protagonist’s Name and Occupation: \$name_occupation Protagonist’s Goal: \$goal Protagonist’s Dire Situation (if the goal is not achieved): \$dire_situation Tell me about a concrete action the protagonist is most likely to take, in order to achieve the goal. <i>Max Sterling, employing their extensive network of contacts and utilizing their advanced tech skills, infiltrates a top-secret meeting of high-ranking Phoenix members, planting surveillance devices to gather valuable intel on the organization’s plans and the whereabouts of its leader. (We denote the model output as \$action1.)</i>
The protagonist tries to perform the following action: \$action1. But the protagonist still fails to achieve the goal for a reason. Tell me about what this reason could be. <i>Despite Max Sterling’s meticulous planning and execution, they are betrayed by a double agent within their own agency, who tips off the Phoenix leaders about the surveillance devices, resulting in the immediate evacuation of the meeting and thwarting Max’s attempt to gather crucial information and track down the organization’s leader. (We denote the model output as \$reason1.)</i>
The protagonist has taken all the following actions: [\$action1]. But it turns out that those actions are not effective in helping the protagonist to achieve the goal, so the goal is still not achieved yet. Tell me about a concrete new action the protagonist is most likely to take next, in order to achieve the goal. <i>Max Sterling decides to track down and approach a former member of “The Phoenix”, who defected years ago but still holds valuable information about the organization and its leader. Using their negotiating skills and persuasive tactics, Max aims to convince the defector to cooperate and provide crucial insider details that will help bring down the terrorist organization once and for all. (We denote the model output as \$action2.)</i>
The protagonist tries to perform the following action: \$action2. But the protagonist still fails to achieve the goal for a reason. Tell me about what this reason could be. <i>Despite Max’s best efforts, the defector is not willing to collaborate, fearing for their own safety and the safety of their loved ones. The defector is haunted by past experiences with “The Phoenix” and their leader, making them reluctant to trust anyone, even someone as skilled and determined as Max. (We denote the model output as \$reason2.)</i>
The protagonist has taken all the following actions: [\$action1, \$action2]. But it turns out ... achieve the goal. (similar to the prompt used for \$action2) <i>Realizing that their previous actions have not yielded the desired results, Max Sterling decides to gain access to The Phoenix’s encrypted communication channels, using their expertise in cryptography and hacking to unravel the organization’s hidden communications and identify potential weak links within the group who could provide vital information about the leader’s location. (We denote the model output as \$action3.)</i>

Table 2: The prompts we use in subsection 3.2 Outline Planning (highlighted in **bold**), and an example of LLM’s corresponding outputs (highlighted in *italic*). All the variables shaped as \$variable are obtained in subsection 3.1, as reflected in Table 1.

pendix B. The corresponding full story is presented in Appendix C.

4 Evaluation

Human participant evaluation has been generally held as the gold-standard evaluation technique for creative language generation tasks such as story generation (Celikyilmaz et al., 2020; Caglayan et al., 2020; van der Lee et al., 2021). To extensively evaluate our method, we design three human participant studies. In the first, participants are asked to judge the goodness of the stories we generate and compare stories generated by our technique against those generated by the baseline method. The comparisons are based on multiple evaluation criteria. In the second, we perform ablations on our system and compare our full system against the ablations. Ablation studies help us understand whether different parts of our system are significant to the solution. In the third, participants are requested to examine the story outline to help determine if the internal workflow is running properly as we expected.

All human participant studies are performed using the Prolific crowdsourcing platform². These studies have been approved by our institution’s Institutional Review Board (IRB). We qualify participants by first asking them a screening question at the beginning of the survey, and then verifying answers manually to disregard answers provided by those who fail the screening. We require participants to be physically located in the U.S. and to speak English as a first language. For each human participant study, we source a distinct set of participants (no overlap) to avoid potential bias in annotations that could occur from participating in related studies in the past. For each study, we measure the average inter-annotator agreement using Fleiss’s kappa (Fleiss, 1971).

We choose ChatGPT (OpenAI, 2022) (gpt-3.5-turbo-0613) as the LLM we use in the implementation. As mentioned in section 3, our method does not rely on any external story corpora. Suspenseful stories can be generated with our method in a fully zero-shot way from ChatGPT. We start every ChatGPT dialogue session with “You are a creative storyteller.”. When generating anything other than stories (e.g., interim information such as the protagonist’s goal and actions), to discourage ChatGPT from

generating excessively long responses that are usually not quite useful when serving as inputs to subsequent generation stages, we append “Use no more than $[n]$ sentences in your answer.” (n can be three, four, or five) to the prompt when necessary.

4.1 Comparison Against Baseline Method

In this study, we seek to understand whether our method has achieved improvement in terms of story suspense as well as other typical story evaluation criteria. We choose the following method as the baseline to compare with: directly asking ChatGPT for a suspenseful story with straightforward prompts. To make it a fair comparison, we control that the stories are generated conditioned on the same background information (the protagonist’s name, occupation, and goal) used in our method’s generation process. The prompt we use for generating the baseline stories is: “Story Background: \$name_occupation. \$goal. Write a full suspenseful story based on the story background.” (variables \$name_occupation and \$goal are obtained from subsection 3.1).

We recruited 90 participants to evaluate 30 pairs of stories, with each individual evaluating 10 pairs of stories, and each pair of stories was evaluated by 30 participants. Each pair of stories comprises one story generated by our method and one story generated by the baseline method. Both stories are randomly selected from a corresponding large collection of stories generated with either our method or the baseline method.

We ask human participants to make a pair-wise comparison and assess the goodness of pairs of suspenseful stories from the following dimensions:

- **Suspense:** “Which story makes you feel a higher level of suspense?”;
- **Novelty:** “Which story do you find to be more novel?”;
- **Enjoyment:** “Which story is more enjoyable to read?”;
- **Logical Sense:** “Which story logically makes more sense?”;
- **Naturalness:** “Which story is more likely to have been written by a human?”.

Results are presented in Table 3 where we show the percentage of times stories generated from each method are preferred for each evaluation metric. Participants’ preferences with a fair average inter-annotator agreement show that compared to the baseline, our method significantly improves the

²<https://www.prolific.co/>